

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 3.1: Statistics I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Statistics I
Driving Question	DRIVING QUESTION: How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Is the Road Really Well Served? Data and Where It Comes From	We observed the County Road Safety Officer's claim that the average distance between crossing points on the 8,000 m stretch of Thika Road is about 400 m (21 crossing points), and that this is true. We also saw learners crossing away from the crossings at our school gate, and we read the log of 12 near-misses kept by Mzee Otieno and other matatu drivers, who say most happen in one stretch.	We learned that data are facts, figures or observations collected about something, and that statistics is collecting, organising, representing and interpreting data to make decisions. We learned three sources of data: surveys (asking), observation (watching, counting and measuring) and records (information already written down). We learned that an undesignated crossing is a pedestrian crossing away from a zebra crossing or footbridge.	We can explain that the officer's average of 400 m may be true but incomplete, because one number may not show whether the crossing points are evenly spaced or where pedestrians are at risk. To test the claim we need to collect our own data on the gaps between crossing points and the pedestrians crossing away from them, and we will keep adding to the Driving Question Board and our model as we learn.
Lesson 2 Counting the Road: Tallies and Frequency Tables	We tallied the 20 gaps between the 21 designated crossing points on the 8,000 m stretch of Thika Road and the pedestrians crossing at undesignated spots in each gap (182 in total). Our frequency tables showed that gap lengths of 100 m to 300 m are common (14 of the 20 gaps), while six gaps are between 600 m and 1,100 m, and six gaps had pedestrian counts of 15 to 31.	Tally marks (in bundles of five) help us collect data, and a frequency distribution table for ungrouped data organises it so that we can check the total (20 gaps) and see which values are common or unusual.	The officer's 400 m average is true but incomplete. Our frequency table shows that most gaps are short while a few are very long, which suggests that a single number hides where pedestrians may be at risk. We need more tools, such as mean, median and mode, to describe the data fairly.
Lesson 3 Mean, mode and median: which one tells the truth?	We calculated the mean, mode and median of the 20 gaps between crossing points on an 8,000 m stretch of Thika Road (mean 400 m, median 250 m, mode 200 m, range	Mean = sum of (f x x) divided by sum of f; median is the middle value of the ordered data (the mean of the 10th and 11th values for 20 values); mode is the most frequent	The officer's claim of a 400 m average is true but incomplete: a few very long gaps pull the mean above the median (250 m) and mode (200 m), so one average hides

	1,000 m) and of the pedestrian counts per gap (mean 9.1, median 4, mode 3, range 30). 14 of the 20 gaps are 300 m or shorter, and six gaps are 600 m to 1,100 m.	value; range is highest minus lowest. The mean is affected by very large values, while the median and mode are not.	the fact that most gaps are short and a few are very long. We still need to find where the long gaps are and how many pedestrians cross there.
Lesson 4 Grouping the Gaps into Classes	We observed that the 20 gaps on the 8,000 m stretch of Thika Road, which have 11 different values from 100 m to 1,100 m, can be grouped into six classes of width 200 m: 0 to <200 (5 gaps), 200 to <400 (9 gaps), 400 to <600 (0 gaps), 600 to <800 (2 gaps), 800 to <1000 (1 gap) and 1000 to <1200 (3 gaps), with a total of 20.	We learned how to choose a class width using the range, write classes with the lower boundary included and the upper excluded (for example 200 to <400), tally each value into exactly one class, check that the frequencies add up to 20 and identify the modal class as the class with the highest frequency, which is 200 to <400.	The grouped table shows that 14 of the 20 gaps are under 400 m while six gaps are 600 m or longer, and no gap lies in 400 to <600 where the officer's mean of 400 m falls. So the officer's claim is true but incomplete: the modal class shows the most common gap but cannot show where along Thika Road the long gaps and the risk are.
Lesson 5 Mean and median from grouped data: what happens to the average when we group the gaps?	We used the grouped Thika Road gap table for the 20-gap sample (classes 0 to <200 ... 1000 to <1200, frequencies 5, 9, 0, 2, 1, 3) to calculate class midpoints (100, 300, 500, 700, 900, 1100), $f \times \text{midpoint}$ (sum 8,800) and cumulative frequencies (5, 14, 14, 16, 17, 20). We found an estimated mean of 440 m and a grouped median of 311 m, and compared them with the ungrouped mean (400 m) and median (250 m).	For grouped data, the mean is estimated using class midpoints: $\text{sum of } (f \times \text{midpoint}) \text{ divided by sum of } f$. The median is found with cumulative frequency and the formula $L + ((n/2 - cf)/f) \times w$. Grouping loses the exact values, so grouped results are estimates that differ from ungrouped results.	The grouped mean (440 m) and median (311 m) differ from 400 m and 250 m because midpoints stand in for the real gaps and the formula assumes values are spread evenly in a class. Ms Halima Abdi's 400 m mean is true but incomplete: none of these averages shows that six gaps of 600 m to 1,100 m sit in one 5,250 m stretch near the school gate, so we need graphs to show where pedestrians are at risk.
Lesson 6 Histograms and frequency density: seeing where the long gaps are on Thika Road	We drew a histogram of the 20 gaps on the 8,000 m stretch of Thika Road with equal 200 m classes (frequencies 5, 9, 0, 2, 1, 3). Then we merged three classes into 400 to <1000 and found that bar height by frequency was misleading. Frequency density gave 2.5, 4.5, 0.5 and 1.5.	A histogram has touching bars on a continuous axis, and bar area represents frequency. Frequency density = frequency / class width (in units of 100 m). With unequal class widths, the tallest bar is the one with the highest frequency density, not the highest frequency.	The histogram shows that most gaps are short (14 of 20 are 300 m or shorter), while a few long gaps of 600 m to 1,100 m lie to the right. These long gaps pull the mean up to 400 m and sit in the long stretch G9-G14 near the school gate. So the officer's mean is true but incomplete, and a graph shows where pedestrians are at risk.
Lesson 7 Frequency Polygons and Comparing Roads: Same Mean, Different Shape	We drew the Thika Road frequency polygon (20 gaps, classes of width 200 m) by plotting (100,5), (300,9), (500,0), (700,2), (900,1), (1100,3), closed at (-100,0) and (1300,0), and drew the Mombasa Road (Embakasi) polygon with frequencies 4, 8, 5, 2, 1, 0 on the same axes, by hand and with a spreadsheet.	A frequency polygon plots (midpoint, frequency), joins the points with straight lines and is closed on the axis at both ends. Putting two polygons on the same axes lets us compare their shape and spread. The grouped means are similar (Thika Road 440 m, Mombasa Road 380 m) and both have the modal class 200 to <400.	The officer's claim is true but incomplete. Similar averages can hide different shapes: Thika Road has a second small rise at 1000 to <1200 m and long gaps, while Mombasa Road is a single hump with no gap of 1000 m or more. One average cannot show where pedestrians are at risk, so we use the polygon together with the median, mode and the long stretch G9-G14. This is one morning's sample and should be repeated.
Lesson 8 Interpreting,	What we observed: On the 8,000 m stretch of Thika Road (20-gap sample), the mean	What we learned: A mean is a useful summary but it can be pulled up by a few	What we can now explain: The officer's claim that the average gap is about 400 m

Recommending and Presenting Findings: Where Is the Road Really Unsafe?	gap is 400 m but the median is 250 m and the mode is 200 m. 14 of 20 gaps are 300 m or shorter, while six gaps of 600 m to 1,100 m (G9-G14) form a 5,250 m long stretch including the school gate. In this stretch 140 of 182 undesignated crossers (77%) and 9 of 12 near-misses were recorded. The Mombasa Road polygon has a similar mean (380 m) but no gap of 1000 m or more.	very long gaps and hide where people are at risk. Median, mode, grouped tables, histograms and frequency polygons show spread and shape, and together they reveal the danger. Statistics helps us check claims, make informed decisions and communicate with the community. A one-morning sample must be repeated before we are certain.	is true but incomplete. Because six very long gaps cluster in one stretch near the school gate, the mean looks reassuring while most undesignated crossings and near-misses occur there. We recommend extra crossing points (for example a zebra crossing or footbridge near the school gate and in the 1000 to <1200 m gaps) and a repeat count, and we presented this evidence to the school administration, parents and community leaders.
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ATTRIBUTION AND LICENSING

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